

PART A

A1. Project Title: AquaGuard cloud Water Quality Reporter with Sensor Analytics and public Alerts.

Main User Action: The user submits water quality sensor data to the cloud and views water quality alerts.

A2. User Request: Send the latest water quality sensor reading to the cloud and display the water quality status and public alerts.

Part B - Network Journey: phone to cloud:

Message path: phone $\rightarrow$ Wi-Fi / 4G / 5G $\rightarrow$ DNS $\rightarrow$ HTTPS / TLS $\rightarrow$ TCP / IP $\rightarrow$ Router $\rightarrow$ ISP $\rightarrow$ Internet $\rightarrow$ cloud Endpoint $\rightarrow$ API Gateway $\rightarrow$ Application Server $\rightarrow$ Database $\rightarrow$ Analytics $\rightarrow$ Service $\rightarrow$ Alert Service $\rightarrow$ user Response.

Technical Translation Table:

* The water quality request travels securely from the user's mobile device to the cloud through multiple networking layers. Each network component performs a specific function to ensure reliable, secure, and fast communication. The table below

explains the role of each component, possible failures and the cloud protection mechanisms used.

Layer / Component	What it does	possible Failure	Cloud protection
DNS	Finds Server Ip address	DNS failure	Backup DNS
HTTPS / TLS	Encrypts data	Certificate error	SSL / TLS Certificates
TCP / IP	Transfers data reliably	packet loss	Retransmission
Router / ISP	Routes packets	Network Congestion	Multiple network paths
API Gateway	Receives API requests	Heavy traffic	load balancing & auto Scaling

part c - Cloud Landing Zone:

The AquaGuard System uses cloud services to securely process, store and analyze water quality data received from sensors. Each cloud component performs a specific function to improve reliability, and scalability and real-time public alert of delivery.

Cloud Mud	Component Choice	Service Model	Justification
User Login	Authentication Service	PaaS	Secure user access
Backend Logic	Cloud Application Server	PaaS	Processing of sensor data
Database	Managed SQL / NoSQL Database	PaaS	Stores water quality records
File Storage	Cloud Storage	PaaS	Stores reports and sensor logs
Monitoring	Cloud Monitoring Dashboard	SaaS	Tracks system performance and alerts

part D - API Contract:

* Endpoint: /api/waterquality/report

* HTTP Method: POST

* Request Data: Sensor ID, pH, Turbidity, Temperature, Timestamp, Location

* Response Data: Status, water Quality Level, Alert Message.

* Authentication: JWT Token / API Key

* Error Case: Invalid sensor data or unauthorized access return HTTP 401/400.

part E - Elasticity, Scalability and Measured Service:

E₁. peak Requests / Minute: $7200 \div 10 = 720$ requests/minute

E₂. Required Capacity (25% spare): $720 \times 1.25 = 900$ request/minute

E₃. Minimum Instances: $900 \div 180 = 5$ instances

E₄. Elastic Design Decision: Enable automatic Scaling of API Servers when sensor traffic increases.

E₅. Dashboard Metrics:

- * Request Count

- * API Response Time (Latency)

part F - AI / Google Research Evidence:

- * Search used: "cloud auto scaling for IoT sensor applications"

- * Tool used: ChatGPT / Google

- * Useful Finding: Auto Scaling automatically adds cloud servers during peak traffic.

- * Applied in project: Used auto scaling to ensure Aqueduct can process thousands of sensor reports without downtime.